Dyeing Faults and Remedies | Common Problems in Dyeing and Their Remedies

Dyeing is a process of coloring fibers, yarns, or fabrics with either natural or synthetic dyes. Many known-unknown faults occur during dyeing operation.

Dyeing Faults

Major dyeing faults which occur during are mentioned below:

1. Uneven Dyeing
2. Batch to Batch Shade variation
3. Patchy dyeing effect
4. Roll to roll variation or Meter to Meter variation
5. Crease mark
6. Dye spot
7. Wrinkle mark
8. Softener Mark

Uneven Dyeing:

Causes:

- Uneven pretreatment (uneven scouring & bleaching).
- Improper color dosing.
- Using dyes of high fixation property.
- Uneven heat-setting in case of synthetic fibers.
- Lack of control on dyeing m/c

Remedies:

- By ensuring even pretreatment.
- By ensuring even heat-setting in case of synthetic fibers.
- Proper dosing of dyes and chemicals.
- Proper controlling of dyeing m/c

Batch to Batch Shade Variation:

Causes:

- Fluctuation of Temperature.
- Improper dosing time of dyes & chemicals.
- Batch to batch weight variation of dyes and chemicals.
- Dyes lot variation.
- Improper reel speed, pump speed, liquor ratio.
- Improper pretreatment.

Remedies:

- Use standard dyes and chemicals.
- Maintain the same liquor ratio.
- Follow the standard pretreatment procedure.
- Maintain the same dyeing cycle.
- Identical dyeing procedure should be followed for the same depth of the Shade.
- Make sure that the operators add the right bulk chemicals at the same time and temperature in the process.
- The pH, hardness and sodium carbonate content of supply water should check daily.

Patchy Dyeing Effect:

Causes:

- Entanglement of fabric.
- Faulty injection of alkali.
- Improper addition of color.
- Due to hardness of water.
- Due to improper salt addition.
- Dye migration during intermediate dyeing.
- Uneven heat in the machine, etc.

Remedies:

- By ensuring proper pretreatment.
- Proper dosing of dyes and chemicals.
- Heat should be same throughout the dye liquor.
- Proper salt addition.

Roll to Roll Variation or Meter to Meter Variation:

Causes:

- Poor migration property of dyes.
- Improper dyes solubility.
- Hardness of water.
- Faulty m/c speed, etc

Remedies:

- Use standard dyes and chemicals.
- Proper m/c speed.
- Use of soft water.

Crease Mark:

Causes:

- Poor opening of the fabric rope
- Shock cooling of synthetic material
- If pump pressure & reel speed is not equal
- Due to high speed m/c running

Remedies:

- Maintaining proper reel sped & pump speed.
- Lower rate rising and cooling the temperature
- Reducing the m/c load
- Higher liquor ratio

Dye Spot:

Causes:

- Improper Dissolving of dye particle in bath.
- Improper Dissolving of caustic soda particle in bath.

Remedies:

- By proper dissolving of dyes & chemicals
- By passing the dissolved dyestuff through a fine stainless steel mesh strainer, so that the large un-dissolved particles are removed.

Wrinkle Mark:

Causes:

- Poor opening of the fabric rope
- Shock cooling of synthetic material
- High temperature entanglement of the fabric

Remedies:

- Maintaining proper reel sped & pump speed.
- Lower rate rising and cooling the temperature
- Higher liquor ratio

Softener Mark:

Causes:

- Improper mixing of the Softener.
- Improper running time of the fabric during application of softener.
- Entanglement of the fabric during application of softener

Remedies:

- Maintaining proper reel sped & pump speed.
- Proper Mixing of the softener before addition.
- Prevent the entanglement of the fabric during application of softener

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